

Communication systems

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ELECTRONICS AND COMMUNICATIONS SYSTEMS

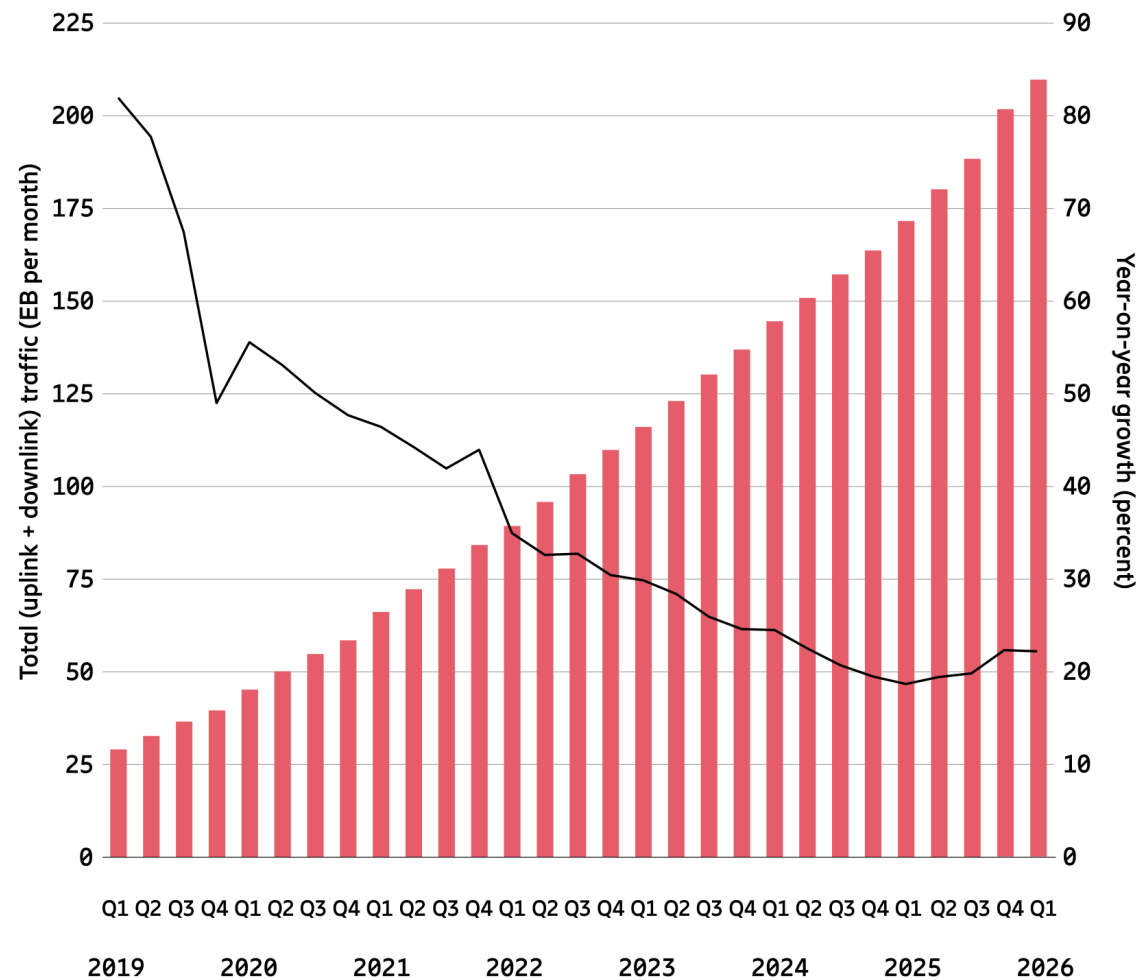
COMPUTER ENGINEERING

Growing Demand, Limited Resources

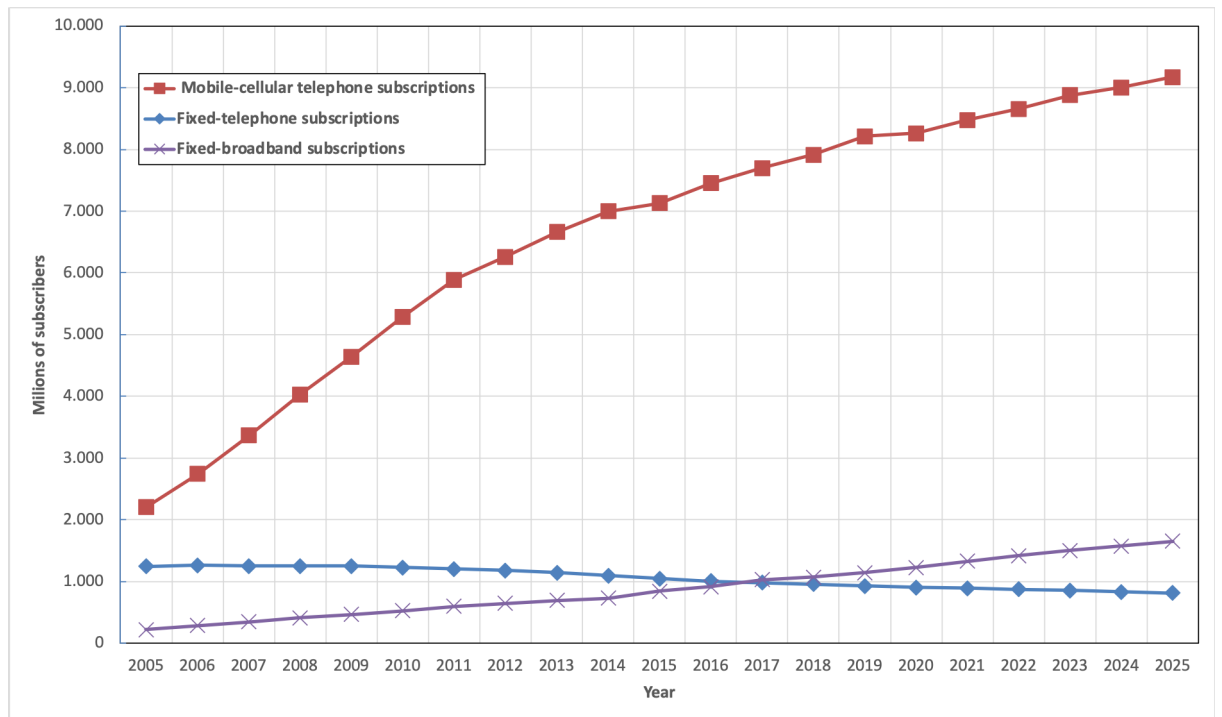
- **Why are communication systems so important?**
- **Connectivity has become an essential infrastructure** for people, industry, transportation and public services.
- The number of connected devices and the amount of exchanged data continue to grow.
- However, the **radio spectrum is a finite and shared resource.**
- *Challenge.* Communication systems must transmit information
 - reliably,
 - at high data rates,
 - using limited bandwidth, power and computational resources.

Mobile traffic growth 2019-2026 (Ericsson)

1 exabyte (EB) = 10^{18} bytes



Evolution of Global Telecommunication Subscriptions by Service (ITU)



Summary of the topics covered in this course

- Intro
- Analog Communications
- Digital Communications
- Wireless propagation channel
- Multi-carrier modulations
- Diversity and MIMO: foundations of modern wireless systems

Introduction

The radio spectrum

UNITED STATES FREQUENCY ALLOCATIONS THE RADIO SPECTRUM

RADIO SERVICES COLOR LEGEND

AERONAUTICAL MOBILE	INTER-SATELLITE	RADIO ASTRONOMY
AERONAUTICAL MOBILE SATELLITE	LAND MOBILE	RADIO DETERMINATION SATELLITE
AERONAUTICAL RADIO NAVIGATION	LAND MOBILE SATELLITE	RADIO LOCATION
AMATEUR	MARITIME MOBILE	RADIO LOCATION SATELLITE
AMATEUR SATELLITE	MARITIME MOBILE SATELLITE	RADIO NAVIGATION
BROADCASTING	MARITIME RADIO NAVIGATION	RADIO NAVIGATION SATELLITE
BROADCASTING SATELLITE	METEOROLOGICAL AND	SPACE OPERATION
EARTH EXPLORATION SATELLITE	METEOROLOGICAL SATELLITE	SPACE RESEARCH
FIXED	MOBILE	STANDARD FREQUENCY AND TIME SIGNAL
FIXED SATELLITE	MOBILE SATELLITE	STANDARD FREQUENCY AND TIME SIGNAL SATELLITE

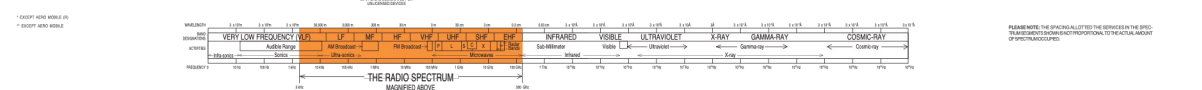
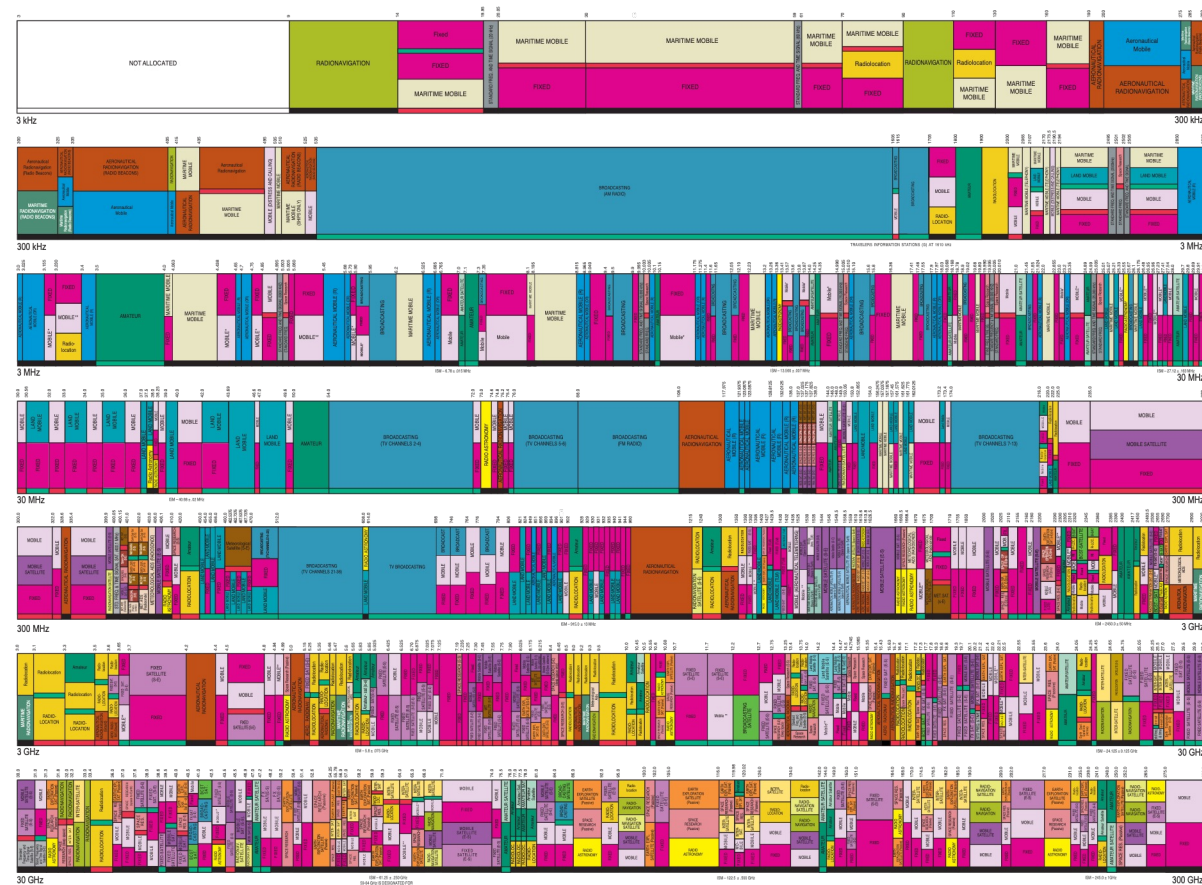
ACTIVITY CODE

GOVERNMENT EXCLUSIVE	GOVERNMENT/GOVERNMENT SHARED
NON-GOVERNMENT EXCLUSIVE	

ALLOCATION USAGE DESIGNATION

SERVICE	EXAMPLE	DESCRIPTION
Primary	FIXED	Capital Letters
Secondary	Mobile	1st Capital with lower case letters

This chart is a graphic representation of the Table of Frequency Allocations used by the FCC and NTIS. As such, it does not constitute a legal document. It is intended to provide a visual summary of the frequency allocations. It is not intended to be used as a legal document. It is intended to provide a visual summary of the frequency allocations. It is not intended to be used as a legal document.

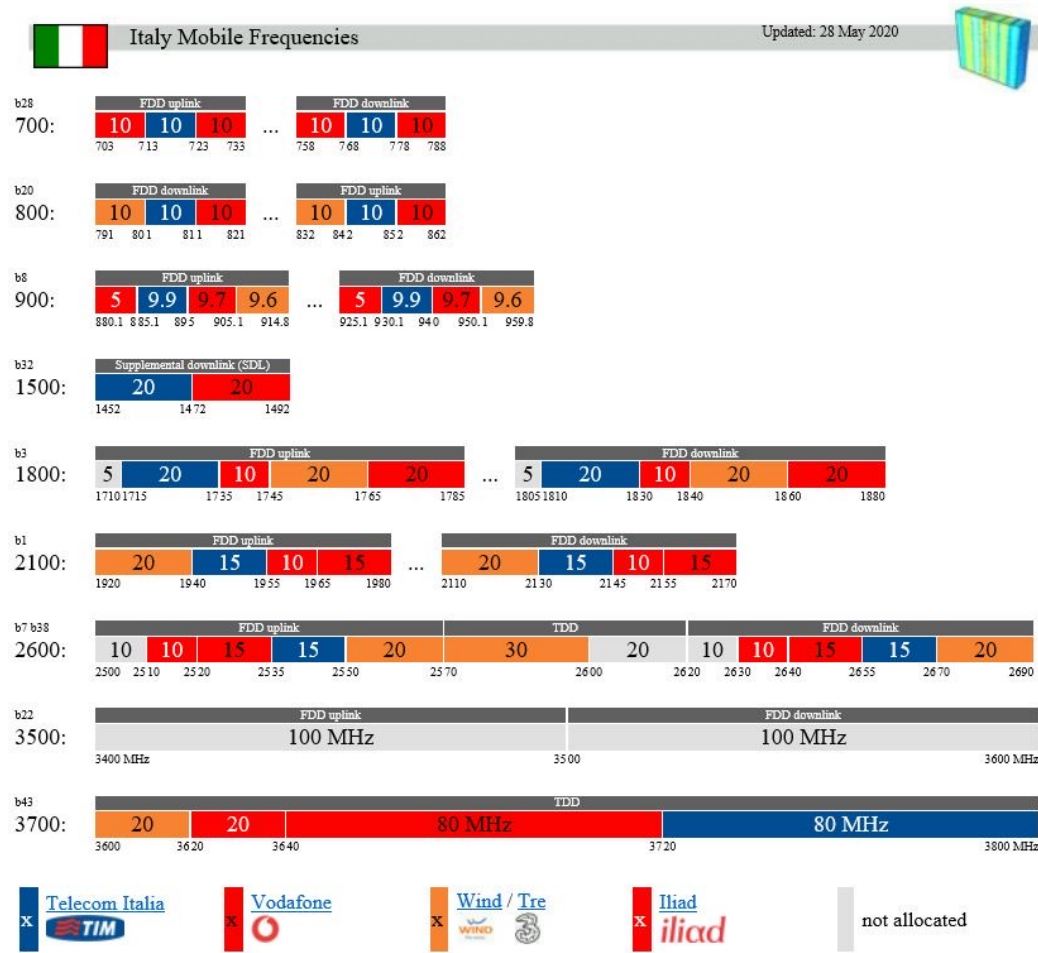


PLEASE NOTE: THE FREQUENCY ALLOCATIONS SHOWN IN THIS CHART ARE SUBJECT TO CHANGE WITHOUT NOTICE. THE ACTUAL ALLOCATIONS MAY VARY FROM THE INFORMATION SHOWN IN THIS CHART.

Wave propagation

Frequency band	Frequency range	Wavelength	Services	Propagation
LF	30 – 300 kHz	$10^4 - 10^3$ m	Radio clock, navigation (LORAN), military (navy)	Ground wave
MF	0.3 – 3 MHz	$10^3 - 10^2$ m	AM radio (520–1600 kHz), radio beacons	Ground wave, sky wave
HF	3 – 30 MHz	$10^2 - 10$ m	Aviation communications, over-the-horizon radar, amateur radio	Sky wave
VHF	30 – 300 MHz	$10 - 1$ m	Avionic communications, FM radio (88 – 108 MHz), DVB-T (RAI@ 177.5 MHz)	Space wave (line-of-sight)
UHF	0.3 – 3 GHz	$1 - 10^{-1}$ m	DVB-T: 470–694 MHz; cellular: 700 MHz–2.6 GHz; GPS: 1.575 GHz; Wi-Fi: 2.4 GHz	
SHF	3 – 30 GHz	$10^{-1} - 10^{-2}$ m	5G mid-band: 3.4–3.8 GHz; Wi-Fi: 5 and 6 GHz; satellite and radar	
EHF	30 – 300 GHz	$10^{-2} - 10^{-3}$ m	5G mmWave; radar, SatCom and future sub-THz systems	

Radio frequency allocation LTE and 5G in Italy



FREQUENZA	PARTECIPANTE	IMPORTO	STATO
700 MHz blocco riservato	ILIAD ITALIA S.P.A.	€ 676.472.792,00	AGGIUDICATO
700 MHz blocco generico	VODAFONE ITALIA S.P.A.	€ 345.000.000,00	AGGIUDICATO
700 MHz blocco generico	TELECOM ITALIA S.P.A.	€ 340.100.000,00	AGGIUDICATO
700 MHz blocco generico	TELECOM ITALIA S.P.A.	€ 340.100.000,00	AGGIUDICATO
700 MHz blocco generico	VODAFONE ITALIA S.P.A.	€ 338.236.396,00	AGGIUDICATO
3700 MHz blocco specifico (80 MHz)	TELECOM ITALIA S.P.A.	€ 1.694.000.000,00	AGGIUDICATO
3700 MHz blocco generico (80 MHz)	VODAFONE ITALIA S.P.A.	€ 1.685.000.000,00	AGGIUDICATO
3700 MHz blocco generico (20 MHz)	WIND TRE S.P.A.	€ 483.920.000,00	AGGIUDICATO
3700 MHz blocco generico (20 MHz)	ILIAD ITALIA S.P.A.	€ 483.900.000,00	AGGIUDICATO
26 GHz blocco generico	TELECOM ITALIA S.P.A.	€ 33.020.000,00	AGGIUDICATO
26 GHz blocco generico	ILIAD ITALIA S.P.A.	€ 32.900.000,00	AGGIUDICATO
26 GHz blocco generico	FASTWEB S.P.A.	€ 32.600.000,00	AGGIUDICATO
26 GHz blocco generico	WIND TRE S.P.A.	€ 32.586.535,00	AGGIUDICATO
26 GHz blocco generico	VODAFONE ITALIA S.P.A.	€ 32.586.535,00	AGGIUDICATO
TOTALE GENERALE		€ 6.550.422.258,00	

Analog Communications

From information signals to radio signals

- Information signals, such as speech and audio, are typically generated at baseband,

$$m(t) \Leftrightarrow M(f), \text{ concentrated around } f = 0$$

- Wireless transmission requires the signal to be moved to an appropriate radio-frequency band.

$$\text{baseband signal} \xrightarrow{\text{modulation}} \text{passband radio signal}$$

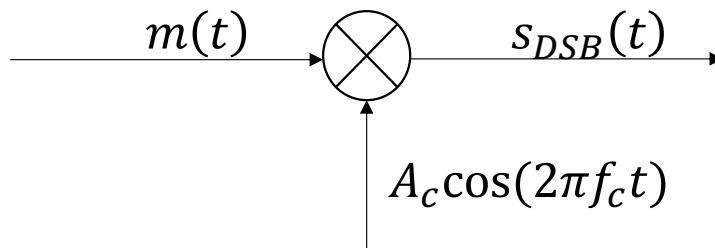
Modulation allows us to:

- translate baseband information to a suitable radio-frequency band;
- enable efficient radiation and reception with practical antennas;
- allow multiple services and users to share the radio spectrum.

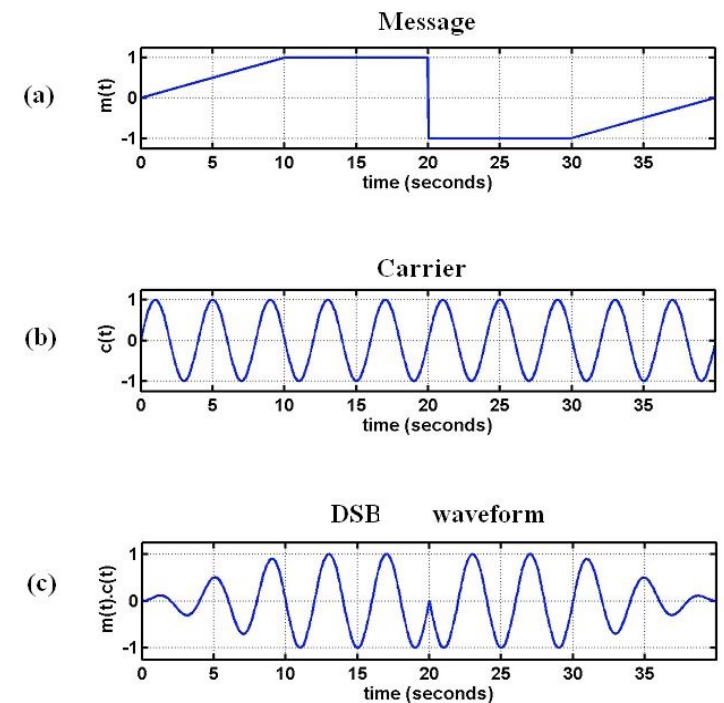
The simplest form of analog modulation: DSB

- A baseband message $m(t)$ can be translated to a radio-frequency band by multiplying it by a sinusoidal carrier.

$$s_{DSB}(t) = A_c m(t) \cos(2\pi f_c t)$$



- Amplitude modulation*: The message $m(t)$ controls the amplitude and sign of the carrier.

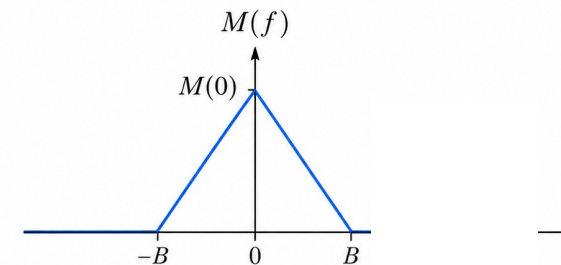


DSB Modulation in the Frequency Domain

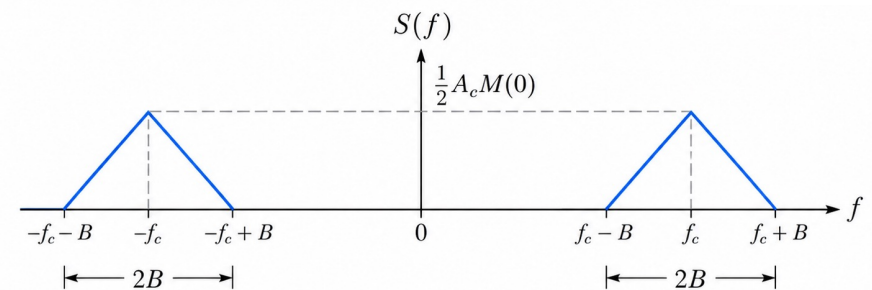
- Multiplication by a sinusoidal carrier creates two shifted copies of the baseband spectrum around f_c and $-f_c$

$$\cos(2\pi f_c t) \Leftrightarrow \frac{1}{2} [\delta(f - f_c) + \delta(f + f_c)]$$

$$S_{DSB}(f) = \frac{A_c}{2} [M(f - f_c) + M(f + f_c)]$$



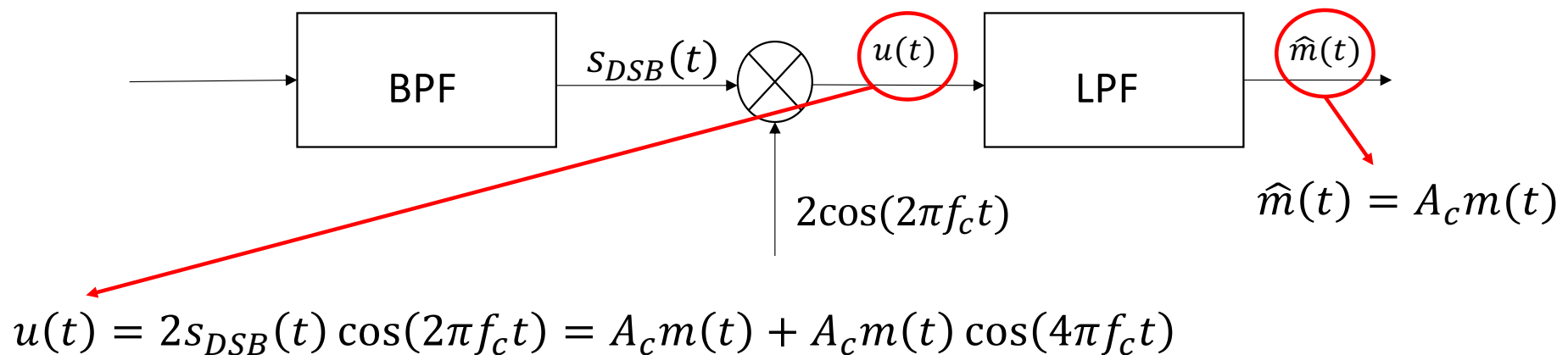
(a) Baseband message spec



(b) DSB signal spectrum

DSB coherent detection

- Neglecting for the moment the effect of the noise and of the propagation channel, recovery of $m(t)$ from $s_{DSB}(t)$ is possible with coherent detection.
- Coherent: the received signal is multiplied by a locally generated carrier with the same frequency and phase.



From DSB to Quadrature Modulation

- Several amplitude-modulation schemes can be obtained by modifying how the carrier and the sidebands are transmitted.
- Rather than reviewing all analog amplitude-modulation schemes, we focus on *quadrature modulation*.
- Can two independent messages share the same frequency band without interfering?
 - The answer is **yes**: by modulating two orthogonal carriers.
- Analog quadrature modulation is especially important because:
 - it allows two independent baseband signals to occupy the same frequency band;
 - it naturally introduces the in-phase and quadrature components;
 - the same principle underlies digital QAM and many modern communication systems.

Quadrature carriers as a transmission resource

- Two carriers at the same frequency and with a 90° ($\pi/2$ radians) phase difference provide two separable transmission dimensions:

$$\cos(2\pi f_c t) \text{ and } -\sin(2\pi f_c t)$$

- Two independent baseband signals $m_I(t)$ and $m_Q(t)$ can be transmitted in the same frequency band

$$s_{QM}(t) = A_c m_I(t) \cos(2\pi f_c t) - A_c m_Q(t) \sin(2\pi f_c t)$$

$m_I(t)$ and $m_Q(t)$ are the *in-phase* and *quadrature* components.

- Both components occupy the same frequency band, but coherent detection can separate them by exploiting the quadrature carriers.

Orthogonality as a Transmission Resource

- Orthogonality is not limited to quadrature carriers at the same frequency. More generally, two signals $x_k(t)$ and $x_\ell(t)$ are *orthogonal* over an interval T if

$$\langle x_k, x_\ell \rangle = \int_{t_0}^{t_0+T} x_k(t) x_\ell^*(t) dt = 0$$

- Sine and cosine carriers at the same frequency are orthogonal over any integer number k of carrier periods: $T = kn = n/f_c$

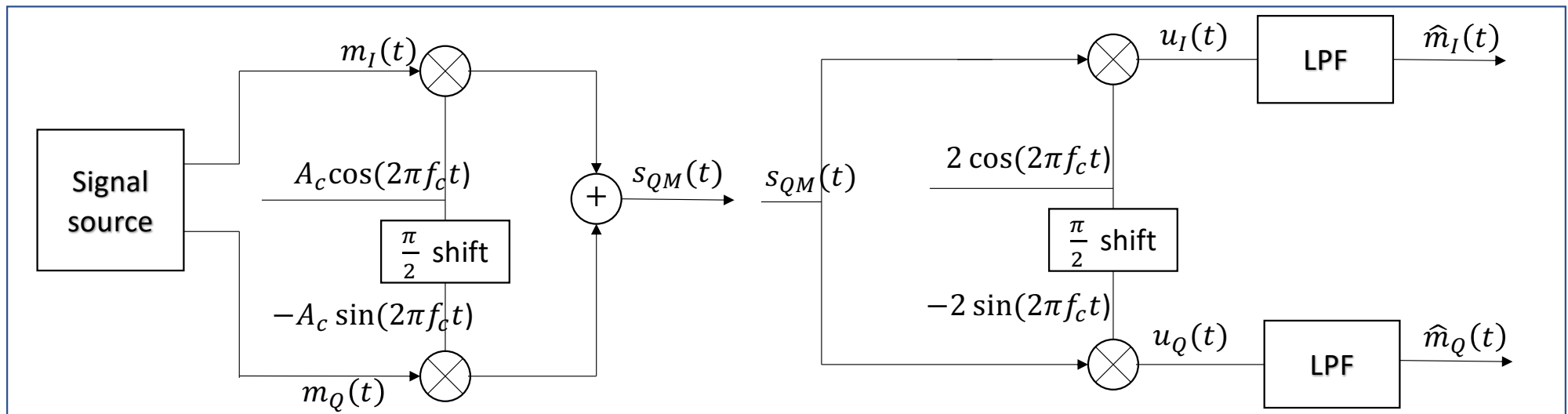
$$\int_{t_0}^{t_0+T} \cos(2\pi f_c t) \sin(2\pi f_c t) dt = 0$$

- Orthogonal signals can *overlap in time and frequency* and still be separated by an appropriate receiver.
 - **QM**: quadrature sine and cosine carriers; the I/Q components are separated by coherent mixing and low-pass filtering.
 - **OFDM**: orthogonality between subcarriers at different frequencies.

Quadrature Modulation: Coherent Receiver

- At the receiver, coherent mixing with the same carriers shifts the desired component to baseband, while the cross-component and the remaining high-frequency terms are centered around $2f_c$.
- Low-pass filtering separates and recovers the two baseband components:

$$\hat{m}_I(t) = A_c m_I(t) \text{ and } \hat{m}_Q(t) = A_c m_Q(t)$$



From I/Q components to a Complex Baseband Signal

- The quadrature-modulated signal is described by two real baseband components:

$$s(t) = A_c m_I(t) \cos(2\pi f_c t) - A_c m_Q(t) \sin(2\pi f_c t).$$

- These two components can be combined into one complex signal:

$$\tilde{s}(t) = A_c m_I(t) + j A_c m_Q(t)$$

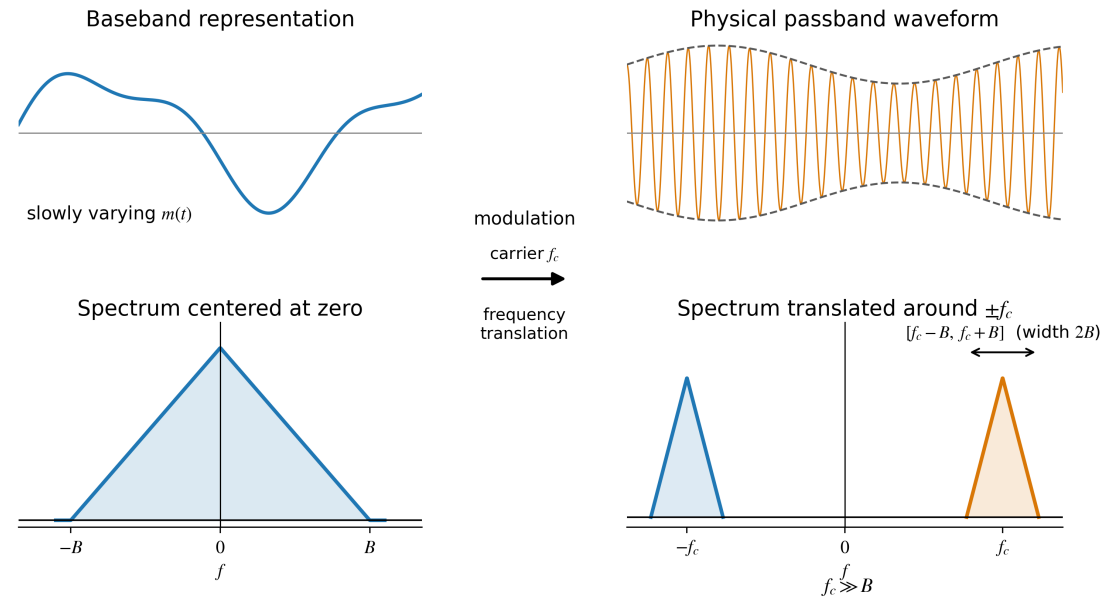
- The physical passband waveform is recovered as

$$s(t) = \text{Re}\{\tilde{s}(t) e^{j2\pi f_c t}\}.$$

- The complex representation is *a mathematical tool*; the transmitted waveform *remains real*.

Passband signals

- The vast majority of communication systems are *passband* systems.
- A passband signal $s(t)$ has its spectrum concentrated around a non-zero carrier frequency f_c .
- In most communication systems, the signal is also narrowband, $f_c \gg B$.



Complex envelope of a passband signal

- Any passband signal $s(t)$ can be represented as

$$s(t) = \text{Re}\{\tilde{s}(t)e^{j2\pi f_c t}\} = s_I(t) \cos(2\pi f_c t) - s_Q(t) \sin(2\pi f_c t)$$

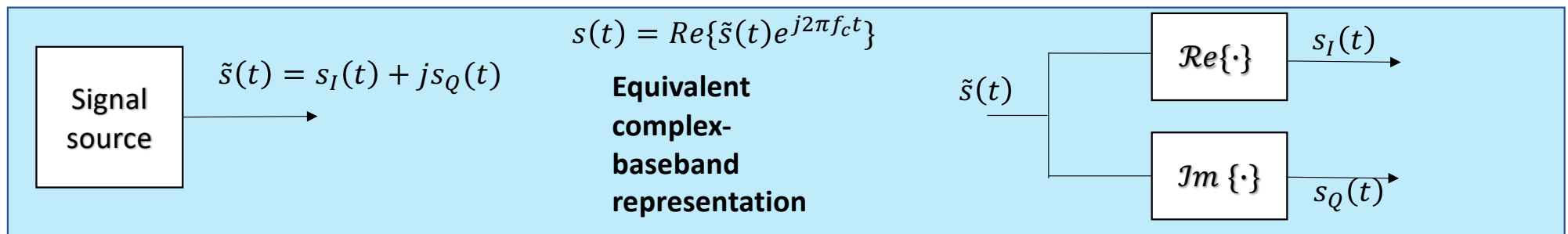
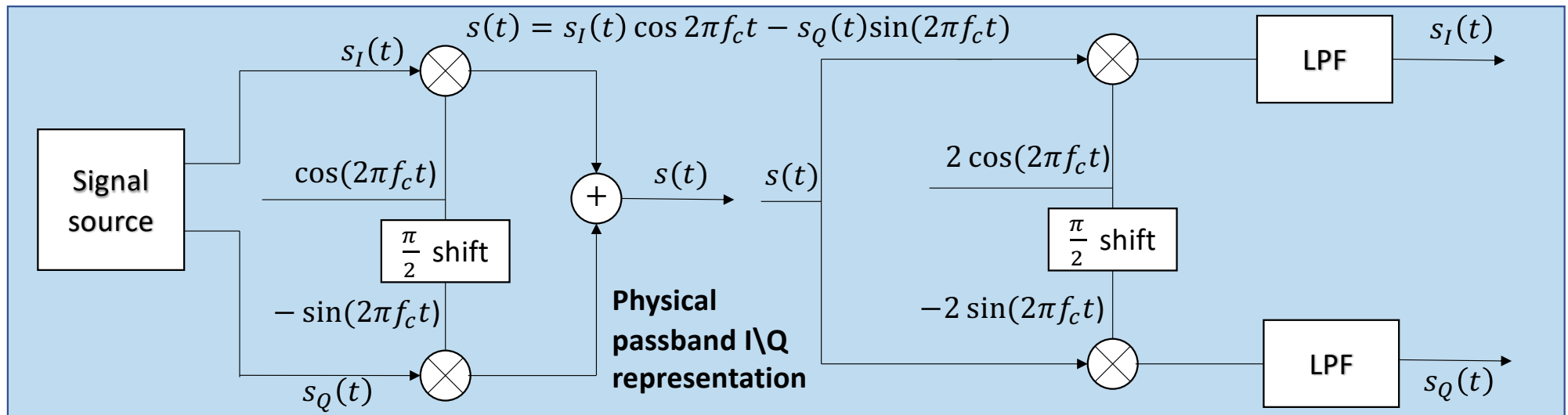
where $\tilde{s}(t) = s_I(t) + js_Q(t)$ is the *complex envelope* of the signal with $s_I(t) = \text{Re}\{\tilde{s}(t)\}$ and $s_Q(t) = \text{Im}\{\tilde{s}(t)\}$ the in-phase and quadrature components.

- Complex envelope for known modulated signals
 - $\tilde{s}_{DSB}(t) = A_c m(t); s_I(t) = A_c m(t), s_Q(t) = 0.$
 - $\tilde{s}_{QM}(t) = A_c m_I(t) + jA_c m_Q(t); s_I(t) = A_c m_I(t), s_Q(t) = A_c m_Q(t).$
 - The carrier amplitude is often normalized to $A_c = 1$ and omitted from the notation.

Complex envelope of a passband signal

- The complex envelope is an equivalent baseband representation of a passband signal.
- Employing the baseband equivalent has several benefits:
 - A baseband model is simpler to study, since it removes the effects of the carrier frequency from the signal model.
 - A baseband model can be numerically simulated with much lower computation complexity than a passband model because the bandwidth and, as a consequence, the sampling rate is much lower.
 - A baseband model is often the basis for a digital implementation of a passband communications system.

Passband vs. equivalent baseband model



Carrier Mismatch in Complex Baseband representation

- In a practical receiver, the phase ϕ_{LO} and the frequency f_{LO} of the local oscillator may not be exactly equal to the phase ϕ_c and frequency f_c of the received carrier.

$$\Delta f_{off} = f_c - f_{LO} \text{ and } \Delta\phi = \phi_c - \phi_{LO}$$

- The *received complex envelope* takes this form

$$\tilde{v}_{off}(t) = \tilde{v}(t)e^{j(2\pi\Delta f_{off}t + \Delta\phi)}$$

- In complex baseband, a phase offset causes a fixed rotation, whereas a frequency offset causes a progressive rotation.

From Amplitude Modulation to Angle Modulation

- A passband signal can always be represented as

$$s(t) = A(t) \cos(2\pi f_c t + \phi_m(t))$$

- Accordingly, information can be conveyed by varying:

- the amplitude $A(t)$;
- the information-bearing phase $\phi_m(t)$;

- Since the instantaneous frequency is the time derivative of the total phase,

$$f_i(t) = \frac{1}{2\pi} \frac{d}{dt} (2\pi f_c t + \phi_m(t)) = f_c + \frac{1}{2\pi} \frac{d}{dt} \phi_m(t),$$

varying the phase also produces a time-varying instantaneous frequency.

- Angle modulation includes phase modulation and *frequency modulation*.

Frequency modulation (FM)

- In frequency modulation, the *instantaneous frequency deviation*

$$\Delta f_i(t) = f_i(t) - f_c = \frac{1}{2\pi} \frac{d}{dt} \phi_m(t) = k_f m(t)$$

is linearly proportional to the message.

- Since it is $d\phi_m(t)/dt = 2\pi k_f m(t)$, it follows that the information-bearing phase term is the integral of the message:

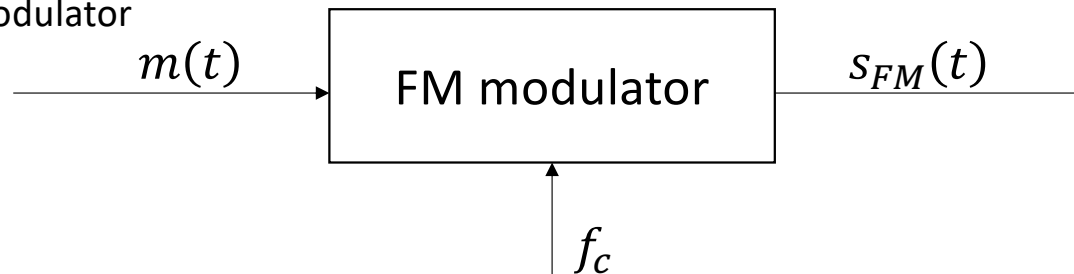
$$\phi_m(t) = 2\pi \int_{-\infty}^t k_f m(\tau) d\tau$$

- The passband FM modulated signal is

$$s_{FM}(t) = A_c \cos (2\pi f_c t + 2\pi k_f \int_{-\infty}^t m(\tau) d\tau)$$

FM Complex Envelope and Main Parameters

Passband FM modulator



- The complex envelope of a FM signal is

$$\tilde{s}_{FM}(t) = A_c e^{j2\pi k_f \int_{-\infty}^t m(\tau) d\tau} = A_c e^{j\phi_m(t)}$$

- Since it is $|\tilde{s}_{FM}(t)| = A_c$, FM is a *constant-envelope* modulation
- Maximum frequency deviation is defined as

$$\Delta f = \max_t \{|\Delta f_i(t)|\} = k_f \max_t \{|m(t)|\}$$

- If the modulating signal has bandwidth B , the modulation index is the dimensionless parameter

$$m_f = \frac{\Delta f}{B}$$

FM signal with a modulating sinusoid

- To study the FM spectrum, we first consider a single-tone message: $m(t)$ is a sinusoid of amplitude V_m and frequency f_m

$$m(t) = V_m \cos(2\pi f_m t)$$

- The FM signal is

$$\begin{aligned} s_{FM}(t) &= A_c \cos \left(2\pi f_c t + 2\pi k_f \int_{-\infty}^t V_m \cos(2\pi f_m \tau) d\tau \right) \\ &= A_c \cos \left(2\pi f_c t + 2\pi k_f V_m \frac{\sin(2\pi f_m t)}{2\pi f_m} \right) \\ &= A_c \cos(2\pi f_c t + m_f \sin(2\pi f_m t)) \end{aligned}$$

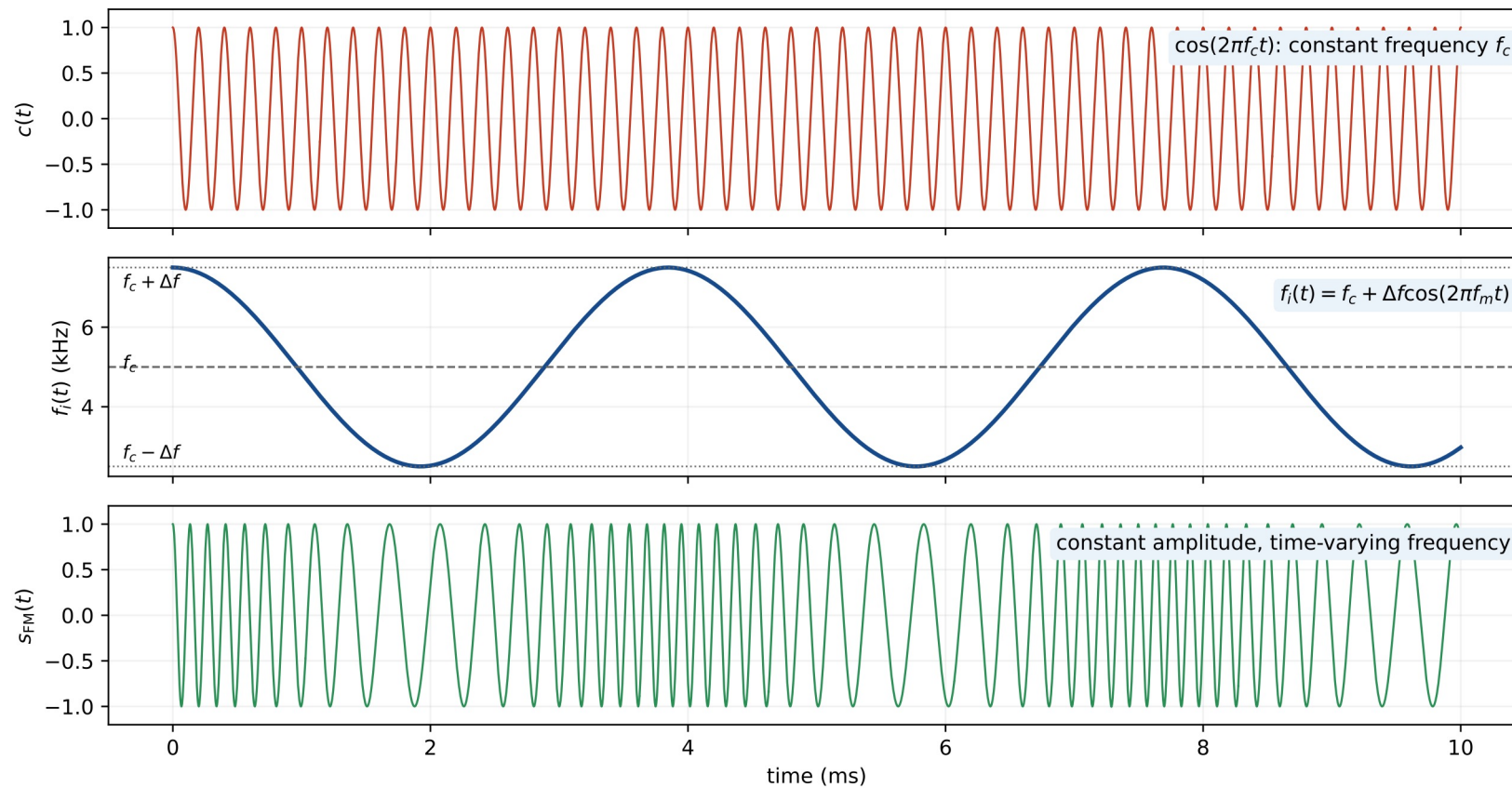
$m_f = \frac{\Delta f}{B}$
 $= \frac{k_f V_m}{f_m} \Delta f$
 $= k_f V_m, B = f_m$

- The complex envelope is

$$\tilde{s}_{FM}(t) = A_c e^{jm_f \sin(2\pi f_m t)}$$

Time-Domain Interpretation of FM

Frequency Modulation: carrier, instantaneous frequency, and FM waveform



FM signal spectrum

- For a sinusoidal message, the complex envelope is periodic of period $T_m = 1/f_m$ and can be expanded into a Fourier series

$$\tilde{s}_{FM}(t) = A_c \sum_n S_n e^{j2\pi n f_m t}$$

where the coefficients S_n are Bessel functions of the first kind

$$\begin{aligned} S_n &= \frac{1}{T_m} \int_{-T_m/2}^{T_m/2} e^{jm_f \sin(2\pi f_m t)} e^{-j2\pi n f_m t} dt \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{j(m_f \sin\theta - n\theta)} d\theta = J_n(m_f) \\ \tilde{s}_{FM}(t) &= A_c \sum_n J_n(m_f) e^{j2\pi n f_m t} \end{aligned}$$

FM signal spectrum

- The passband signal is

$$s_{FM}(t) = A_c \sum_n J_n(m_f) \cos(2\pi(f_c + nf_m)t)$$

$$= \sum_n A_n \cos(2\pi f_n t)$$

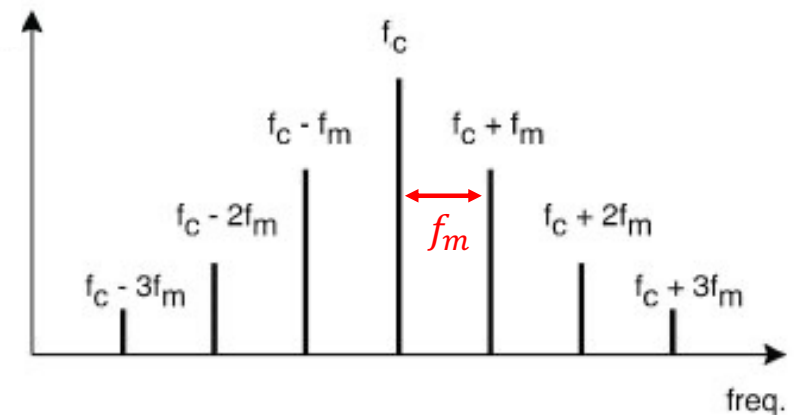
where $A_n = A_c J_n(m_f)$ and $f_n = f_c + nf_m$.

- To interpret the spectrum, we use the property

$$|J_n(m_f)| = |J_{-n}(m_f)|$$

which shows that the upper and lower sidebands have the same magnitude.

- As m_f increases, higher-order sidebands become significant; since adjacent sidebands are spaced by f_m , the *effective bandwidth* grows roughly as $2(m_f + 1)f_m$



Signal bandwidth B

FM signal spectrum

- For a general message signal, the FM spectrum does not admit a simple closed-form expression.
- A good approximation is the Carson bandwidth rule

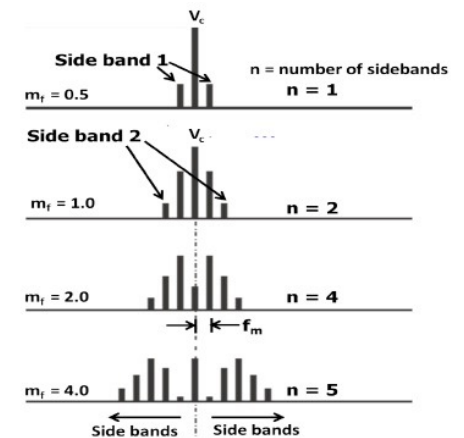
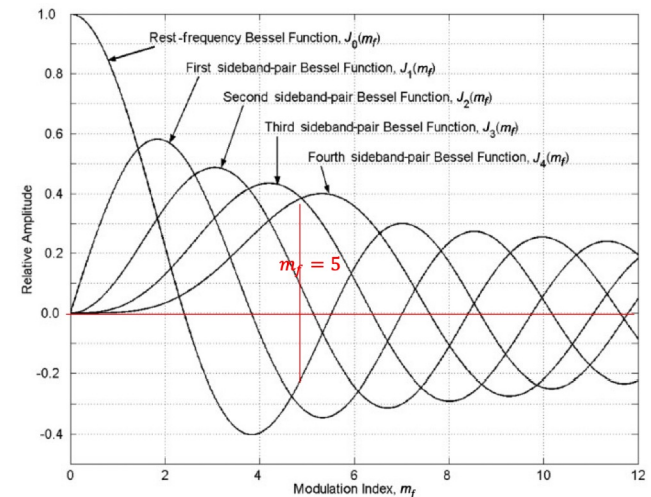
$$B_{FM} \approx 2(m_f + 1)B = 2(\Delta f + B)$$

- An FM signal theoretically has infinitely many sidebands and therefore infinite bandwidth; however, approximately 98% of its power is contained within Carson's bandwidth.

- In commercial mono FM we have

$$B_{FM} \approx 180 \text{ kHz}$$

- $B = 15 \text{ kHz}$ (high quality audio)
- $\Delta f = 75 \text{ kHz}$
- $m_f = 5$



FM: Features and Trade-offs

- Main features:
 - **Constant envelope:** enables the use of efficient nonlinear power amplifiers and allows amplitude limiting at the receiver.
 - **Bandwidth–noise robustness trade-off:** increasing k_f , and hence the maximum frequency deviation Δf , improves noise robustness at the cost of a larger transmission bandwidth.
 - **Capture effect:** in the presence of two FM signals at nearby frequencies, the receiver tends to demodulate the stronger one rather than their sum.
 - **Threshold effect:** FM provides good noise performance above a certain received signal level, but its quality degrades rapidly below this threshold.
- Commercial broadcast FM uses an audio bandwidth of 15 kHz, a maximum deviation of 75 kHz, and a channel spacing of 200 kHz.

FM receiver

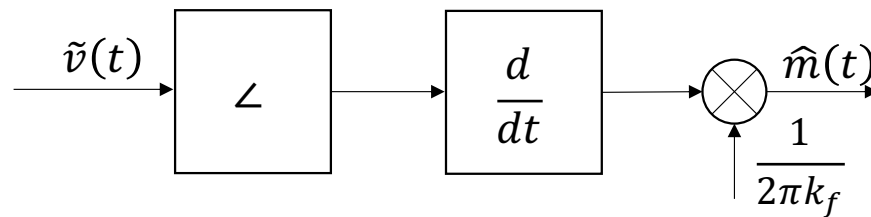
- Neglecting the effect of noise and channel, the complex envelope of the received signal is

$$\tilde{v}(t) = A_c e^{j2\pi k_f \int_{-\infty}^t m(\tau) d\tau}$$

- The modulating signal can be recovered by differentiating the phase of $\tilde{v}(t)$

$$\hat{m}(t) = \frac{1}{2\pi k_f} \frac{d}{dt} \angle \tilde{v}(t)$$

Conceptual FM baseband receiver



Effect of Carrier Phase and Frequency Offsets

- FM demodulation rejects constant phase errors and converts frequency offsets into DC bias.

$$\tilde{v}_{off}(t) = A_c e^{j2\pi k_f \int_{-\infty}^t m(\tau) d\tau} e^{j(2\pi \Delta f_{off} t + \Delta \varphi)}$$

$$\angle \tilde{v}_{off}(t) = 2\pi k_f \int_{-\infty}^t m(\tau) d\tau + 2\pi \Delta f_{off} t + \Delta \varphi$$

$$\frac{d}{dt} \angle \tilde{v}_{off}(t) = 2\pi k_f m(t) + 2\pi \Delta f_{off}$$

$$\hat{m}(t) = \frac{1}{2\pi k_f} \frac{d}{dt} \angle \tilde{v}_{off}(t) = m(t) + \frac{\Delta f_{off}}{k_f}$$

